# SIMATS ENGINEERING
## ASSESSMENT TOOL 3

**COURSE NAME: CSA1610 COURSE CODE: Data Warehousing and Data Mining**

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**COURSE OUTCOME COVERED**

**CO5:** Apply association rule mining techniques to discover interesting relationships and patterns within large datasets

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**QUESTIONS: 05** **Total Marks:50**

Annexure A

**MOCK INTERVIEW**

| ***Criteria*** | Technical Knowledge ***(2.5)*** | Problem Solving ***(2.5)*** | Analytical Thinking ***(1.5)*** | ***Communication(1.5)*** | Confidence ***(1)*** | Response to Questions(1) | ***Total(10)*** |
|---|---|---|---|---|---|---|---|
| *Weightage* | *25%* | 25% | 15% | 15% | 10% | *10%* | *100%* |
| *Q1* | | | | | | | |
| *Q2* | | | | | | | |
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***Code of Conduct:***

*I G.Hima Bindu (192511377) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:*** ***G. Hima Bindu***

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## RUBRICS FOR CALCULATION:

| Criteria | Weightage | Level 4 (Exemplary) | Level 3 (Proficient) | Level 2 (Developing) | Level 1 (Beginning) |
|---|---|---|---|---|---|
| Technical Knowledge | 25 | Demonstrates strong and accurate subject knowledge with in-depth understanding | Shows good knowledge with minor gaps | Basic knowledge with noticeable errors | Lacks fundamental technical knowledge |
| Problem Solving | 25 | Solves problems logically and applies concepts effectively in real scenarios | Applies concepts with moderate accuracy | Limited problem-solving ability; requires guidance | Unable to solve or apply concepts |
| Analytical Thinking | 15 | Analyzes questions critically and provides well-structured, logical answers | Provides reasonable analysis with some structure | Superficial or partially structured responses | No analytical thinking demonstrated |
| Communication | 15 | Communicates clearly, confidently, and professionally with proper terminology | Communicates well with minor hesitation | Communication lacks clarity or confidence | Unable to communicate effectively |
| Confidence | 15 | Displays high confidence, positive attitude, and professional behavior throughout | Shows good confidence with minor issues | Low confidence; inconsistent behavior | Lacks confidence and professionalism |
| Response to Questions | 10 | Responds accurately, promptly, and elaborates with relevant examples | Responds correctly but with limited depth | Partial responses; needs prompting | Unable to respond appropriately |

**Annexure A**

Mock Interview – Sample Questions (5 Questions)

Q1. Dataset: Retail Transactions

| TID | Items |
|---|---|
| T1 | Milk, Bread |
| T2 | Milk, Butter |
| T3 | Bread, Butter |
| T4 | Milk, Bread, Butter |
| T5 | Milk, Eggs |

**Question:**
Based on the dataset, explain how association rule mining can improve sales in a retail store. Provide a practical example.

Q2. Dataset: Transaction Data

| TID | Items |
|---|---|
| T1 | A, B |
| T2 | B, C |
| T3 | A, C |
| T4 | A, B, C |
| T5 | B, C |

**Question:**
Using this dataset, explain the differences between Apriori and FP-Growth algorithms and when each should be used.

Q3. Dataset: Transaction Data

| TID | Items |
|---|---|
| T1 | A, B |
| T2 | A |
| T3 | B |
| T4 | A, B |
| T5 | A |

**Question:**
Explain support and confidence using the dataset and describe their importance in association rule mining.

Q4 . Dataset: Multilevel Data

| TID | Items |
|---|---|
| T1 | Electronics, Mobile |
| T2 | Electronics, Laptop |
| T3 | Electronics, Mobile |
| T4 | Electronics, Mobile |
| T5 | Electronics, Laptop |

**Question:**
Explain multilevel association rules using the dataset and compare them with multidimensional

Q5. Dataset: Large Transaction Sample

| TID | Items |
|---|---|
| T1 | A, B, C |
| T2 | A, B |
| T3 | B, C |
| T4 | A, C |
| T5 | A, B, C |

**Question:**
Discuss challenges in mining large datasets using the given sample and explain how scalable methods address these issues.

Q1. Dataset: Retail Transactions

| TID | Items |
|---|---|
| T1 | Milk, Bread |
| T2 | Milk, Butter |
| T3 | Bread, Butter |
| T4 | Milk, Bread, Butter |
| T5 | Milk, Eggs |

**Question:**
Based on the dataset, explain how association rule mining can improve sales in a retail store. Provide a practical example.

**Association Rule Mining in Retail**

Association rule mining finds relationships between products that are frequently purchased together. Retail stores can use these relationships for cross-selling, product placement, recommendations, and promotions.

From the dataset:

- Milk appears in T1, T2, T4, T5 → **4/5 = 80% support**
- Bread appears in T1, T3, T4 → **3/5 = 60% support**
- Butter appears in T2, T3, T4 → **3/5 = 60% support**
- Milk and Bread occur together in T1 and T4 → **2/5 = 40% support**

For the rule **Milk → Bread**:

**Confidence = Support (Milk, Bread) / Support(Milk)**
= (2/5) / (4/5) = **50%**

This means that among customers who buy Milk, **50% also buy Bread**.

**Practical example:**
The store could place bread close to milk or offer a Milk + Bread combo discount**.** It could also recommend bread to a customer buying milk online. This can increase cross-selling and overall sales.

**Interview point:** Association rules help retailers understand customer buying behaviour rather than relying only on intuition.

Q2. Dataset: Transaction Data

| TID | Items |
|---|---|
| T1 | A, B |
| T2 | B, C |
| T3 | A, C |
| T4 | A, B, C |
| T5 | B, C |

**Question:**
Using this dataset, explain the differences between Apriori and FP-Growth algorithms and when each should be used.

**Apriori vs FP-Growth**

Both **Apriori** and **FP-Growth** are used to find frequent item sets, but they work differently.

### Apriori

Apriori generates candidate item sets and repeatedly scans the database.

For example, from the dataset:

- A occurs in T1, T3, T4 → **3/5 = 60%**
- B occurs in T1, T2, T4, T5 → **4/5 = 80%**
- C occurs in T2, T3, T4, T5 → **4/5 = 80%**
- B,C occurs in T2, T4, T5 → **3/5 = 60%**

Apriori uses the principle that **if an itemset is infrequent, all of its supersets must also be infrequent**. This helps prune unnecessary candidates.

### FP-Growth

FP-Growth does not generate a large number of candidate item sets. Instead, it builds an **FP-tree**, which compresses the transaction database, and then mines frequent patterns from the tree.

**Comparison**

| Feature | Apriori | FP-Growth |
|---|---|---|
| Candidate generation | Yes | No |
| Database scans | Multiple | Usually fewer |
| Memory | Can require more | FP-tree requires memory |
| Speed on large data | Generally slower | Generally faster |
| Implementation | Simpler to understand | More complex |

**When to use:**

- Use Apriori when the dataset is relatively small and simplicity is important.
- Use FP-Growth for large datasets with many transactions, where candidate generation would become expensive.

**Interview Point:**

"Apriori is easier to understand but can be computationally expensive because of candidate generation and repeated database scans. FP-Growth is generally more efficient because it compresses the database into an FP-tree and avoids candidate generation.

Q3. Dataset: Transaction Data

| **TID** | **Items** |
|---|---|
| T1 | A, B |
| T2 | A |
| T3 | B |
| T4 | A, B |
| T5 | A |

**Question:**
Explain support and confidence using the dataset and describe their importance in association rule mining.

## Support and Confidence

The dataset contains 5 transactions:

- A occurs in T1, T2, T4, T5 → **4 transaction**
- B occurs in T1, T3, T4 → **3 transactions**
- A and B occur together in T1 and T4 → **2 transactions**

### Support

Support tells us **how frequently an itemset occurs in the entire dataset**.

For {A, B}:

**Support(A,B) = 2/5 = 40%**

So, 40% of all transactions contain both A and B.

### Confidence

Confidence tells us how often B is purchased when A is purchased.

For the rule **A → B**:

**Confidence(A → B) = Support(A,B) / Support(A)**

= (2/5) / (4/5)
= **50%**

Therefore, **50% of customers who bought A also bought B**.

For comparison, for **B → A**:

Support(A,B) = 2/5
Support(B) = 3/5

Confidence = (2/5) / (3/5) = **66.67%**

### Importance

- **Support** helps identify patterns that occur frequently enough to be useful.
- **Confidence** measures the reliability of an association rule.
- Both help eliminate rules that are either too rare or unreliable.

**Interview point:**

A high confidence rule is not automatically useful; in practice, we may also examine **lift** to determine whether the relationship is genuinely stronger than expected by chance.

Q4 . Dataset: Multilevel Data

| **TID** | **Items** |
|---|---|
| T1 | Electronics, Mobile |
| T2 | Electronics, Laptop |
| T3 | Electronics, Mobile |
| T4 | Electronics, Mobile |
| T5 | Electronics, Laptop |

**Question:**
Explain multilevel association rules using the dataset and compare them with multidimensional

**Multilevel Association Rules vs Multidimensional Association Rules**

Multilevel association rules discover relationships at different levels of abstraction in a concept hierarchy.

In this dataset:

- Level 1: **Electronics**
- Level 2: **Mobile / Laptop**

For example:

**Electronics → Mobile**

Mobile occurs in T1, T3 and T4:

**Support = 3/5 = 60%**

Laptop occurs in T2 and T5:

**Support = 2/5 = 40%**

A multilevel analysis could therefore identify patterns such as:

- Customers frequently buy Electronics
- Within Electronics, Mobile is more frequently purchased than Laptop

The key idea is that the same data can be analysed at different levels of detail**.**

### Multilevel vs Multidimensional

| Feature | Multilevel | Multidimensional |
|---|---|---|
| Focus | Different abstraction levels | Different attributes/dimensions |
| Example | Electronics → Mobile | Age → Income → Product |
| Hierarchy | Usually uses concept hierarchy | Uses multiple attributes |
| Purpose | Analyse patterns at different granularities | Analyse relationships across attributes |

**Interview Point:**

"Multilevel association rules look at the same type of information at different levels, such as Electronics → Mobile. Multidimensional association rules combine different dimensions or attributes, such as Customer Age, Location, Income, and Product.

Q5. Dataset: Large Transaction Sample

| TID | Items |
|---|---|
| T1 | A, B, C |
| T2 | A, B |
| T3 | B, C |
| T4 | A, C |
| T5 | A, B, C |

**Question:**
Discuss challenges in mining large datasets using the given sample and explain how scalable methods address these issues.

Challenges in Mining Large Datasets

Even though the sample contains only 5 transactions, it demonstrates the types of patterns that become challenging when the dataset becomes very large.

For example, the dataset contains combinations such as:

- A,B,C
- A,B

- B,C
- A,C

As the number of items increases, the possible number of item combinations grows rapidly. This creates several challenges.

### Major challenges

1. **Large search space**
   With many items, the number of possible itemsets can become extremely large.
2. **High computational cost**
   Searching through millions or billions of transactions requires significant processing power.
3. **Repeated database scans**
   Algorithms such as basic Apriori may need multiple scans of the database.
4. **Memory limitations**
   Large transaction datasets may not fit entirely into memory.
5. **I/O and processing time**
   Reading and processing huge datasets can become a major bottleneck.

### How scalable methods solve these problem

FP-Growth reduces candidate generation by constructing an FP-tree.

Apriori pruning removes item sets that cannot possibly become frequent.

Partitioning divides a large dataset into smaller portions that can be processed separately.

Sampling can analyse a representative subset when an approximate answer is acceptable.

Parallel and distributed processing allows frameworks such as distributed data-processing systems to divide the workload across multiple machines.

**Interview answer:**
"The main challenge in large-scale association mining is the explosion of possible item combinations and the computational cost of processing huge transaction databases. Scalable algorithms such as FP-Growth, together with pruning, partitioning, sampling, and distributed processing, reduce computation and allow frequent-pattern mining to work efficiently on large datasets."